Number Theory Makeup Midterm

50 minutes • 100 points • Six problems

Grading note. Award method credit after earlier errors. Suggested working times for Problems 1–6 are 7, 7, 7, 7, 6, and 10 minutes, leaving 6 minutes to check. Coverage is based on the midterm review bank.
Instructions. Answer all six problems and show your reasoning. No calculator is needed. You may quote results proved in class unless a proof is requested. Use the spaces provided; clearly label any work continued on the back. Reserve time to check your work.

1. Euclid and a linear congruence — 15 points

(a) Use the Euclidean algorithm to find $\gcd(195, 84)$. (5 points)
Solution. $195 = 2 \cdot 84 + 27$, $84 = 3 \cdot 27 + 3$, $27 = 9 \cdot 3$. The last nonzero remainder is 3, so the gcd is 3.
Credit: Correct divisions: 4; identifying the gcd: 1.

(b) Express this gcd as an integer linear combination of 195 and 84. (5 points)
Solution. $3 = 84 - 3 \cdot 27 = 84 - 3(195 - 2 \cdot 84) = 7 \cdot 84 - 3 \cdot 195$.
Credit: Back substitution: 3; correct coefficients: 2. Accept any valid pair.

(c) Find all solutions of $84x \equiv 6 \pmod{195}$. Give the distinct residues modulo 195 and explain why your list is complete. (5 points)
Solution. Doubling (b) gives $x = 14$ as a solution. Dividing the congruence by 3 gives $28x \equiv 2 \pmod{65}$. Since 28 is coprime to 65, this has one class modulo 65, namely 14. The three classes modulo 195 are $14, 79, 144$.
Credit: A solution: 2; all three classes: 2; completeness: 1.

2. Continued fractions and an inverse — 15 points

Let $D(a) = \left(\begin{smallmatrix} a & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

(a) Find the simple continued fraction of $139/60$ and all its convergents. Use a final digit at least 2. (6 points)
Solution. Euclid gives remainders $19, 3, 1, 0$ and quotients $2, 3, 6, 3$. Thus $139/60 = [2; 3, 6, 3]$, with convergents $2/1, 7/3, 44/19, 139/60$.
Credit: Digits: 2; convergents: 1 each.

(b) Multiply the digit matrices for your expansion. Use the determinant of their product to find the inverse of 60 modulo 139. (9 points)
Solution.

$$D(2)D(3)D(6)D(3) = \begin{pmatrix} 139 & 44 \\ 60 & 19 \end{pmatrix}.$$

The determinant is $(-1)^4 = 1$, hence $139 \cdot 19 - 60 \cdot 44 = 1$. Therefore $60^{-1} \equiv -44 \equiv 95 \pmod{139}$.
Credit: Matrix: 5; determinant/Bézout identity: 2; inverse: 2.

3. Reassembling residues and reducing powers **15 points**

(a) Solve $x \equiv 2 \pmod 5$ and $x \equiv 4 \pmod 7$. Justify uniqueness modulo 35. (7 points)

Solution. Write $x = 2 + 5k$. Modulo 7, $5k \equiv 2$, so $k \equiv 6$. Thus $x \equiv 32 \pmod{35}$. A difference of solutions is divisible by both 5 and 7, hence by 35, and adding 35 preserves both residues.

Credit: Substitution and solution: 4; coprimality and uniqueness: 3.

(b) Find the multiplicative order of 3 modulo 7 and use it to compute $3^{100} \pmod 7$. Show why your order is the least possible. (8 points)

Solution. The successive powers $3^1, \ldots, 3^6$ modulo 7 are $3, 2, 6, 4, 5, 1$. The first occurrence of 1 is at exponent 6, so the order is 6. Since $100 \equiv 4 \pmod 6$, $3^{100} \equiv 3^4 \equiv 4 \pmod 7$.

Credit: Order with minimality justification: 5; exponent reduction and result: 3.

4. Prime factors and divisor sums **15 points**

Here $\tau(n)$ counts positive divisors, $\varphi(n)$ counts units modulo n, and μ is the Möbius function.

(a) Factor 180 and compute $\tau(180)$, $\varphi(180)$, and $\mu(180)$. Show the prime-factor calculations. (6 points)

Solution. $180 = 2^2 3^2 5$. Hence

$$\tau(180) = 3 \cdot 3 \cdot 2 = 18, \qquad \varphi(180) = 180\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) = 48.$$

Since a prime square divides 180, $\mu(180) = 0$.

Credit: Factorization: 1; divisor count: 2; totient: 2; Möbius value: 1.

(b) Suppose f satisfies $\sum_{d \mid n} f(d) = n$ for every positive integer n. Use Möbius inversion to give a formula for $f(n)$ and compute $f(12)$. (9 points)

Solution. Inversion gives

$$f(n) = \sum_{d \mid n} \mu(d)\frac{n}{d}.$$

At $n = 12$, only $d = 1, 2, 3, 6$ contribute, so $f(12) = 12 - 6 - 4 + 2 = 4$.

Credit: Correct inversion formula: 5; nonzero terms: 3; value: 1.

5. Valuations and a 5-adic limit **15 points**

For a prime p, let $v_p(r)$ denote the exponent of p in a nonzero rational r, with $v_p(a/b) = v_p(a) - v_p(b)$. Define $|r|_5 = 5^{-v_5(r)}$ for $r \neq 0$, $|0|_5 = 0$, and $d_5(x, y) = |x - y|_5$.

(a) State Legendre's formula for $v_p(n!)$ and use it to compute $v_3(30!)$. (5 points)
Solution.

$$v_p(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{p^j} \right\rfloor, \qquad v_3(30!) = 10 + 3 + 1 = 14.$$

Terms vanish once $p^j > n$.
Credit: Formula: 2; summands: 2; result: 1.

(b) Compute $|125/14|_5$ and $d_5(1/3, 16/3)$. (5 points)
Solution. Since $v_5(125/14) = 3$, $|125/14|_5 = 1/125$. Also $1/3 - 16/3 = -5$, of valuation 1, so the distance is $1/5$.
Credit: First valuation and norm: 2; difference, valuation, and distance: 3.

(c) Let $S_n = \sum_{j=0}^{n} 5^j$. Find its 5-adic limit in $\mathbb{Q}$ and justify your answer by computing the distance to the limit. (5 points)
Solution. The geometric sum is $S_n = (5^{n+1} - 1)/4$. Thus the limit is $-1/4$, since

$$d_5(S_n, -1/4) = \left| \frac{5^{n+1}}{4} \right|_5 = 5^{-(n+1)} \longrightarrow 0.$$

Credit: Geometric sum and proposed limit: 2; exact distance: 2; convergence: 1.

6. A field of formal pairs **25 points**

Use formal pairs $a + bs$ with $a, b \in \mathbb{F}_7$ and the relation $s^2 = 3$. Addition is coefficientwise; multiplication is distributive with s^2 replaced by 3. You may use the resulting commutative ring laws. All scalar arithmetic in this problem is modulo 7.

(a) List the squares in $\mathbb{F}_7$. Show that $a^2 - 3b^2 \neq 0$ whenever $(a, b) \neq (0, 0)$, and deduce that every nonzero pair has inverse $(a - bs)/(a^2 - 3b^2)$. (11 points)

Solution. The squares are $0, 1, 2, 4$, so 3 is nonsquare. If $b = 0$, then $a \neq 0$ and $a^2 \neq 0$. If $b \neq 0$, vanishing of $a^2 - 3b^2$ would give $(a/b)^2 = 3$, a contradiction. Since $(a + bs)(a - bs) = a^2 - 3b^2$, division by this nonzero scalar supplies an inverse. Thus the ring is a field.

Credit: Squares/nonsquare: 4; denominator proof (both cases): 5; inverse: 2.

(b) Compute $(2 + s)(3 + 2s)$ and $(1 + s)^{-1}$. Check the inverse by multiplication. (8 points)

Solution.

$$(2 + s)(3 + 2s) = 6 + 7s + 2s^2 = 12 = 5.$$

Also $(1 + s)^{-1} = (1 - s)/(1 - 3) = 3(1 - s) = 3 + 4s$. Checking: $(1 + s)(3 + 4s) = 3 + 7s + 4s^2 = 15 = 1$.

Credit: Product: 4; inverse: 2; multiplication check: 2.

(c) Give a basis over $\mathbb{F}_7$, the cardinality, and the characteristic. Explain why this field differs from $\mathbb{Z}/49\mathbb{Z}$. (6 points)

Solution. The basis is $1, s$: formal pairs have unique coefficients in $\mathbb{F}_7$. There are $7^2 = 49$ elements and the characteristic is 7. In $\mathbb{Z}/49\mathbb{Z}$, the characteristic is 49 and the nonzero class 7 has square zero, so that ring is not a field.

Credit: Basis and justification: 2; cardinality: 1; characteristic: 1; valid distinction from the residue ring: 2.

End of paper. Check that you have answered all six problems.